

STUDY OF RECTIFIED WAVEFORMS Using Arduino Uno

VINAY KUMAR K.

This project can be used in the laboratories of industrial training institutes (ITIs), polytechnics, and engineering colleges to study the half-wave and full-wave rectifier waveforms using an Arduino Uno. While electrical power from the mains supply is available as AC, almost all electronic equipment use DC power supply.

The author's prototype is shown in Fig. 1 and circuit diagram of a rectifier using Arduino Uno is shown in Fig. 2. This circuit is built around Arduino Uno, centre-tap transformer X1, two diodes, a dual channel relay module, and a few other components. The circuit provides different types of rectifier outputs, such as full-wave and half-wave with and without filters.

Arduino Uno's digital pins 8 and 7 are connected to input pins IN1 and IN2 of the dual-channel relay module, respectively. Common (COM1) and normally open (NO1) terminals of the relay are connected to transformer's secondary terminal and anode of diode D2, respectively.

NO2 terminal of the relay is connected to positive terminal of electrolytic capacitor C1 and COM2

terminal of the relay is connected to the junction of cathodes of diodes D1 and D2 and resistor R1. Negative

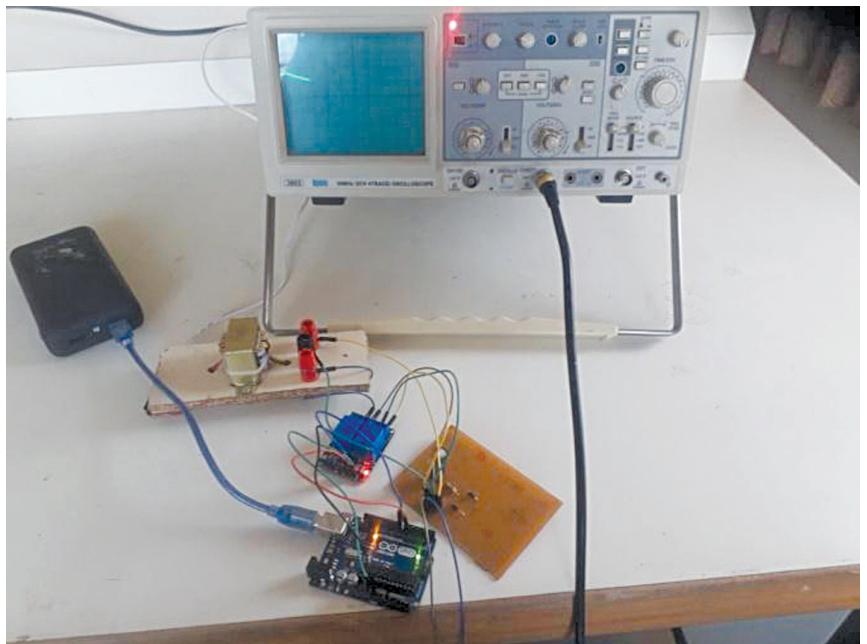


Fig. 1: Author's prototype

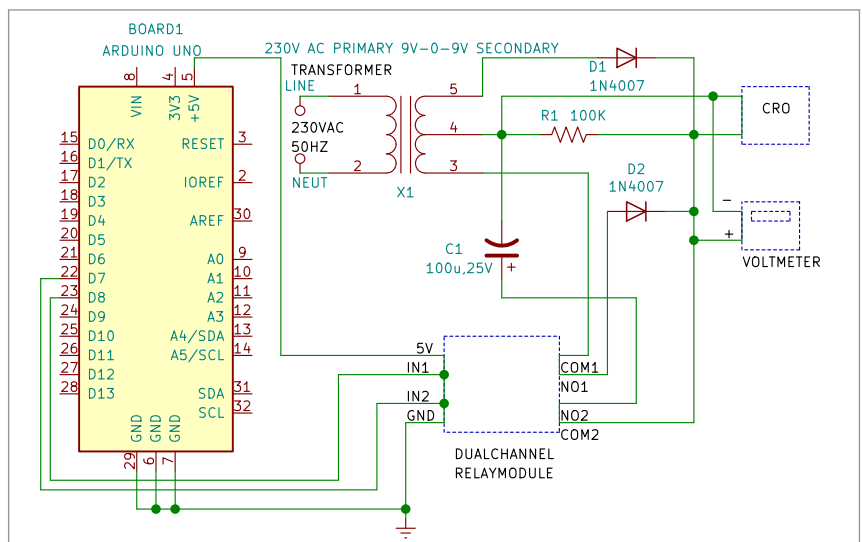


Fig. 2: Circuit diagram

PARTS LIST

Semiconductors:

- Board1 - Arduino Uno board
- D1-D2 - 1N4007 rectifier diode

Resistor (1/4-watt, $\pm 5\%$ carbon):

- R1 - 100-kilo-ohm

Capacitor:

- C1 - 100uF, 25V electrolytic

Miscellaneous:

- X1 - 230V AC primary to 9V-0-9V, 250mA secondary transformer
- Dual-channel relay module (5V DPDT relay module)
- Cathode ray oscilloscope (CRO)
- Voltmeter